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Writing Paper's Threats to Validity

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Outline

- Threats to validity
- Writing the threats to validity section
- Paper review
- Exercise

Organizing (conference paper)

- Title (1000 readers)
- Abstract (4 sentences, 100 readers)
- Introduction (1 page, 100 readers)
- The problem (1 page, 10 readers)
- My idea (2 pages, 10 readers)
 - Proposed approach
- The details (5 pages, 3 readers)
 - Experimental design
 - Results
 - Discussions
 - Threats to validity
- Related work (1-2 pages, 10 readers)
- Conclusions and further work (0.5 pages)

Source: “[How to write a great research paper - Seven simple suggestions](#)” by Simon Peyton Jones, Microsoft Research, Cambridge

Validity Evaluation

- In many engineering papers involving experimentation, we need to provide a discussion on threats to validity of the study
- A fundamental question
 - How valid the results are? → are there any **concerns** that affect the validity of the results?
- Examples
 - Errors or missing steps in the experimental design
 - Inappropriate or lack of measures for evaluating accuracy of a model
 - Data does not represent the general population
 - Your app is designed for doctors but you evaluate your app using students

Four Types of Threats to Validity

■ Internal Validity

- Is there a **causal relationship** between the treatment and the effect/outcome?

■ Construct Validity

- Does the **selected variables reflect** the **construct** of the cause and the effect well?

■ Conclusion Validity

- Threats to conclusion validity affect the **ability to draw the correct conclusion** about relations between the treatment and the outcome

■ External Validity

- Can the results be **generalized** outside of the research scope?

Threats to Internal Validity

■ History

- Different treatments applied to the same object at different times. Are the circumstances the same?

■ Maturation

- The subjects react differently at different times (e.g., tired, bored, more knowledge)

■ Selection

- Concerning the selection of participants/subjects. Is the sample representative for the whole population?

■ Instrumentation

- Are documents used for the experiment correct and inspected?

■ Mortality

- Participants drop out of the experiment

■ Treatment order

- How much the participants know about the treatment?

Threats to Construct Validity

- What is *construct*?
 - From Dictionary:
 - “an idea or theory containing various conceptual elements, typically one considered to be subjective and not based on empirical evidence”
- A construct is explained by one or more explanatory variables
 - E.g., student performance in this course is explained/measured via individual and group assignments → student performance is a construct

Threats to Construct Validity

- Inadequate preoperational explication of constructs
 - A method A is better than B. What does this mean?
- Mono-operation bias
 - The experiment does not sufficiently represent the construct
- Confounding constructs and levels of constructs
 - The difference depends on if the subjects have 1, 3, 5 years of experience
- Interaction of different treatments
 - We cannot conclude whether the effect is due to either of treatments or of a combination of treatments
- Hypothesis guessing
 - The subjects base their behaviors on their guesses about the hypothesis

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Threats to Conclusion Validity

- Low statistical power
 - Low p-value, effect sizes, etc.
- Reliability of measures
 - Are the measures relevant and reliable? Examples of measures: MSE, F1
- Violated assumptions of statistical test
 - Some statistical tests are more robust than others
- Fishing and the error rate
 - Fishing for a specific result
- Reliability of treatment implementation
 - The application of treatments to subjects must be standardized

Threats to External Validity

- Interaction of selection and treatment
 - Sample/data set is not representative of the general population
 - e.g., student subjects may not be representative of doctors
 - Can your results be generalized to the general population?
- Interaction of setting and treatment
 - Material is not representative
 - Toy problems
- Interaction of history and treatment
 - The experiment is conducted in a special time which affects the results

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Writing the Threats to Validity Section

- Papers involving experiments should include a section discussing possible threats to validity
- This section usually appears before conclusion
- You do not have to write all four types of threats to validity
 - Focusing on [internal and external threats to validity](#)
- You can discuss one or several threats under on each type of threats to validity

Examples

Conference paper:
Threats to validation section is
described in less than 1 page (1/4 to
1/2 page usually)

Source: An Nguyen, Bach Le, Vu Nguyen. "Prioritizing automated user interface tests using reinforcement learning". The 15th International Conference on Predictive Models and Data Analytics in Software Engineering (PROMISE 2019). September, 2019. Porto de Galinhas, Brazil

6.3 Threats to validity

One concern of internal validity is whether APFD is appropriate for measuring the performance of the models evaluated. We used this metric because it is widely used for evaluating the performance of test case prioritization methods in previous studies [17][4][5][21]. Thus, the use of APFD helps facilitate comparative assessments of similar methods for test case prioritization. Another concern is the selection of Spectrum as it may not represent typical software systems. Spectrum is a mature application that has gone through many releases and has a number of test cases. It is also open-source and publicly available on Github. Our use of Spectrum along with its data necessary for this study readily available on Github allows future replicated studies to easily use Spectrum and other similar open-source applications.

The use of student groups for code modification, test cases generation and execution might affect the **generalizability** of this study. Student groups may not represent the general population of software project teams. Still, all students are in their senior year and majoring in computer science. Many of them have interned in industry. The processes were established and executed by the students groups independently to simulate a typical process of modifying source code, modifying test cases, and executing test cases in a typical software project. These setups help mitigate the concern of external validity. Additionally, our detailed experiment process can be replicated in a future study.

Examples

Journal paper:

Threats to validation section should be longer, from 1/2 to 2 pages.

This section may be grouped into subsections: internal, external, conclusion and construct validity

Vu Nguyen, Barry Boehm, and LiGuo Huang. "Determining Relevant Training Data for Effort Estimation Using Window-based COCOMO Calibration." *Journal of Systems and Software*, 147, 124-146. 2018

6. Threats to validity

We consider various important threats to validity, which are reported according to four categories, namely, conclusion, internal, construct, and external validity (Wohlin et al., 2012).

6.1. Conclusion validity

Threats to conclusion validity concern "the ability to draw the correct conclusion about relations between the treatment and the outcome" (Wohlin et al., 2012). This study employed the automated calibration procedure that employs several rules to make decisions about outlier elimination and whether to calibrate both constants *A* and *B*, only *A*, or not at all. As discussed in Section 3.2, *F*-tests were used to test the overall significance of a calibrated model with the confidence level of 0.05. These rules may lead to calibrated models different from those produced by performing careful manual calibration and analysis applying the specific knowledge that an organization has about its data. With the specific knowledge about its data and following a manual checking process, an organization may better identify and eliminate truly influential data points. Nonetheless, the rules applied in this study are consistent with the calibration practice recommended for COCOMO and with the procedure employed by Lokan and Mendes in their study of moving windows (Lokan and Mendes, 2009; Lokan and Mendes, 2016). In addition, given numerous calculations and model comparisons applied consistently across many windows and data sets, we believe that this automated calibration procedure is reasonable for comparatively assessing estimation performance of calibrated models across windows.

Another threat to conclusion validity concerns with the consistency of data. As the data were collected and provided by many data vendors, one concern is the lack of consistency in data between projects and between organizations, which may come from vendors' different data collection practices and their various levels of understanding and experience with COCOMO. Moreover, although many cost drivers can be determined objectively, several others such as Analyst Capability (ACAP) and Programmer Capability (PCAP) require subjective judgments to measure their rating levels. There are several ways these concerns are mitigated. First, the published COCOMO II book and the user manual provide detailed model's definition, standards, and guidelines for estimators and data collectors to measure effort, size, and cost drivers. Each cost driver is clearly defined along with detailed criteria and guidelines for measuring its rating levels. Rating levels for many cost drivers are unequivocal. For example, three years of experience in platforms must be mapped to the rating of *High* for the Platform Experience (PLEX) cost driver. Second, most organizations providing the data used COCOMO for estimating projects, and thus, gained much understanding and experience in measuring data for the model. Finally, for the COCOMO data sets, the COCOMO team followed up with the vendors to validate and per-

COCOMO uses 4 to 6-point rating scales, which are not likely sufficient to capture the variation in a cost driver. The defined rating levels may not be refined enough to reflect the true value. For instance, the experience level for Platform Experience (PLEX) between *High* and *Very High* is 3 and 6 years, respectively. But if a team's experience is 5 years, then there is no defined level for this value. A practical method to handle this limitation was to define intermediate levels and extrapolate their rating values. Another issue is that the actual rating value falls outside the defined range of each cost driver, as we discussed in Section 5.2. If this was the case, the estimator had to use either end of the range. Nonetheless, addressing these issues is one of the objectives of the on-going COCOMO III project that the COCOMO team is taking.

6.2. Construct validity

One major threat to construct validity is concerned with whether a study defines measurements appropriately to measure what it purports to measure (Wohlin et al., 2012). In this study, we used three popular estimation accuracy measures, namely MAR, MMRE, and SA. Although MMRE has been criticized as being bias, it is the most widely used measure and has the benchmarking advantage. MAR is an unbiased criterion that has been used effectively in many previous software estimation studies (e.g., (Lokan and Mendes, 2009; Lokan and Mendes, 2016; Kocaguneli et al., 2015)). SA is also unbiased but can serve as a baseline for interpreting estimation accuracy that MAR lacks. Additionally, Cliff's delta effect size was used as an indicator of practical significance in accuracy differences as suggested by Kitchenham et al. (Kitchenham et al., 2017).

6.3. Internal validity

Threats to internal validity affect the causal relationship between the treatment and the outcome of a study (Wohlin et al., 2012). First, there is a concern with selection bias as the data sets are not random samples to represent the entire population of software projects. We used all five organizations that have enough projects instead of randomly sampled sub-data sets for local calibration. Although the projects were collected from 4 countries, most (90%) were conducted by the organizations located in the United States. The large majority of the projects followed heavy-process and model-based software development methodologies like Waterfall as they were completed by 2009 when agile methodologies were just emerging. Process-centric and model-

6.4. External validity

External validity is concerned with the generalizability of the study's results to other software projects (Wohlin et al., 2012). One threat to external validity is whether the study's findings drawn from the nonrandom data sets can be generalized to other projects. COCOMO defines 23 parameters including size to capture many key characteristics for a wide range of projects of different sizes, complexities, domains, and sources. The data sets

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Paper Review

- Paper review is an activity volunteered by researchers in the research community
- Researchers contribute to review as a means of contribution
- Typical steps
 - Read the assigned paper, e.g., in 1-3 hours
 - Write review
 - Give scores for the paper
 - Submit review based on a form

Paper Review

- Typical sections
 - Summary of the paper
 - Provide strengths and weaknesses
 - Provide suggestions/ recommendations
 - Provide scores for different criteria
 - Provide the overall evaluation score

Paper Review

■ Advice

- ❑ Be modest in providing comments
- ❑ Do not be too tough on your criticisms
- ❑ Focus on the paper, not the authors
- ❑ Be sure before you criticize
- ❑ Check your writing
 - When you point out writing issues of the paper, your writing has to be correct
- ❑ Generally, provide reviews as if you were the authors to receive them

Exercise 1

- Read papers related to your topic
 - Point out whether those papers discuss threats to validity
 - If yes, what types of threats to validity are discussed most
 - Provide a summary of your activity
- What is the difference between Internal and External Threats to validity?
- Time: 40 minutes

Exercise 2

- Write the Experiment, Results and Discussion sections for your paper